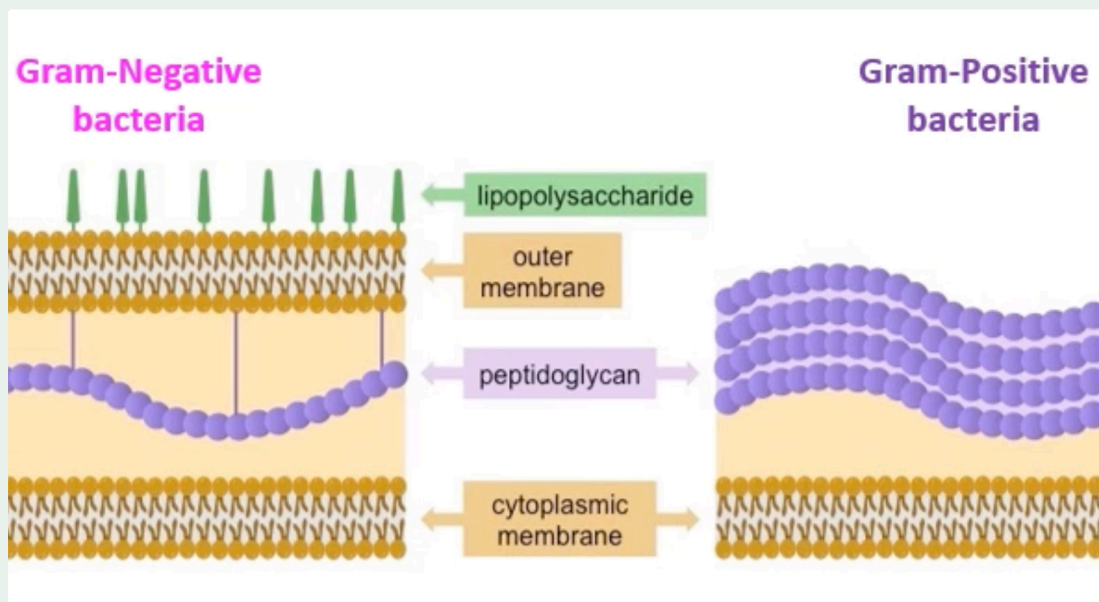


4.4 Diagnostic Bacteriology I

- ▼ Why is diagnosis to find out which particular pathogen is the causative agent important?
 - Because some diseases (i.e pneumonia) can be caused by many different pathogens
 - These pathogens are not all treated in the same way, so it is important to identify the type of pathogen **so we can use appropriate treatments for the patient.**
- ▼ What colour do gram-positive bacteria stain under the gram stain?
 - Purple
- ▼ What colour do gram-negative bacteria stain under the gram stain?
 - Pink
- ▼ Describe the structural difference between gram-positive and gram-negative bacteria.
 - **Gram-positive bacteria** have **one outer membrane** with **thick peptidoglycan** but **gram-negative bacteria** have **two outer membranes** with **thin peptidoglycan**.



▼ Describe the test that is done to distinguish between gram-positive and gram-negative bacteria.

- Bacteria are stained with crystal violet stain
- Bacteria are stained with iodine
- Bacteria are stained with alcohol to wash out crystal violet and iodine
- Bacteria are stained with safranin as the pink stain

▼ Why is it that gram-positive bacteria appear purple and gram-negative bacteria appear pink?

- The thick peptidoglycan in gram-positive bacteria does not allow crystal violet to be washed out and so the bacteria are stained purple
- The thin peptidoglycan in gram-negative bacteria allows crystal violet to be washed out and so the bacteria are stained pink by safranin instead

▼ What are the two types of bacterial shapes?

- Cocci (round spheres)
- Bacilli (rod shaped cylinders)

▼ What is diplococci?

- A pair of two cocci bacteria

▼ What is staphylococci?

- A group of multiple cocci bacteria.

▼ What is streptococci?

- A group of multiple cocci bacteria in a chain arrangement.

▼ Describe the two ways a microscope helps in diagnostic bacteriology.

- It helps us identify the **colour that the bacteria stains in crystal violet solution**
- It helps us identify the **shape of the bacteria**

▼ How does knowing the shape and the stain colour of bacteria help us as clinicians?

- It can help us to identify the bacterial strain that is causing a particular infection

▼ What is a hemolysin?

- **Enzymes that are released by bacteria that can damage red blood cells.**

▼ What is alpha-hemolysis?

- Partial hemolysis
 - green, opaque zone

▼ What is beta-hemolysis?

- Complete hemolysis
 - clear zone

▼ What is gamma-hemolysis?

- No hemolysis

▼ What is the hemolysis test used for?

- Distinguishing between **gram-positive bacteria** such as staphylococcus.

▼ How is the lactose test used to distinguish between bacteria?

- **Bile salts, crystal violet and lactose** are present on the agar plate
- **Bacteria produce lactic acid** when they ferment lactate during their growth
- This results in the **production of pink colonies** as the pH is lower.

▼ What is the lactose test used for?

- Distinguishing between **gram-negative bacteria**.

▼ What is the catalase test used for?

- To **distinguish between streptococci and staphylococci**

▼ What does catalase do?

- It is an enzyme that breaks down hydrogen peroxide into hydrogen and oxygen.

▼ What happens in the catalase test?

- **Staphylococci produce catalase** and hence **bubbles**
- **Streptococci do not produce catalase** and hence **no bubbles are produced**

▼ What is the coagulase test used for?

- To distinguish between staphylococcus aureus and other staphylococci

▼ What does coagulase do?

- Catalyses the **conversion of prothrombin into thrombin**
- **Thrombin** is then used to convert **fibrinogen into fibrin**
- **Factor 8A** then crosslinks **fibrin to form fibrin clot**

- ▼ What happens in the coagulase test?
- Staphylococcus aureus produces clumps (fibrin clot)
 - Other staphylococci produce no clumps